

Overview

A computation fluid dynamics (CFD) analysis was performed to evaluate the drag characteristics of various exterior designs for a robotic fish. The objective was to compare external geometries to optimize hydrodynamic flow around the exterior while fulfilling internal packaging requirements for the actuators.

Geometry

The model was validated using a body with a known drag coefficient. A benchmark was generated from a model used in the professor's biomimicry research. Multiple external body geometries were evaluated to understand the effect of body shape on hydrodynamic performance.

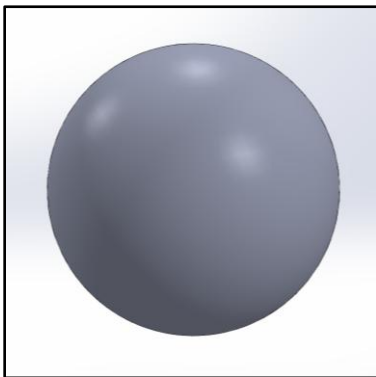


Figure 1. Validation Case

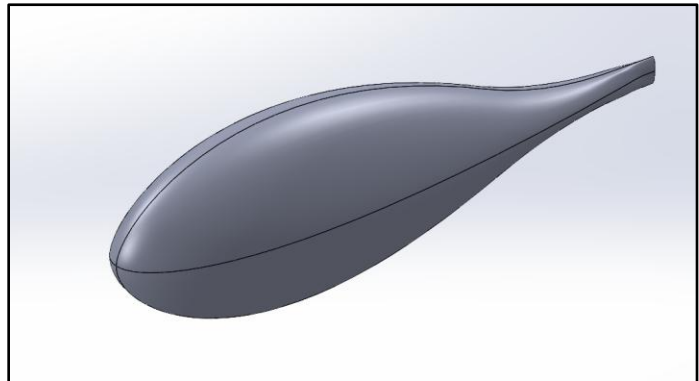


Figure 2. Benchmark

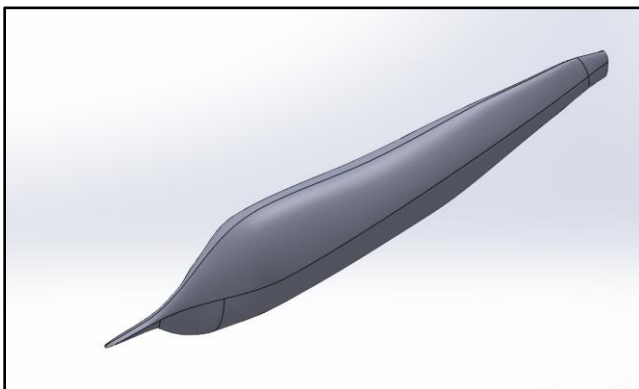


Figure 3. Sail Fish

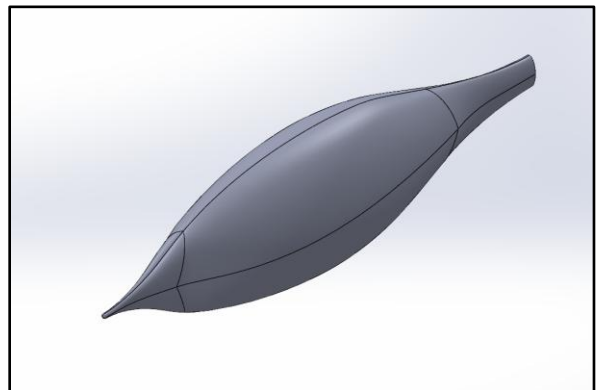


Figure 4. Narwhal

Simulation

The robotic fish geometry was imported into ANSYS Fluent for external flow evaluation. The simulation was configured to identify drag behavior of different body iterations by comparing drag coefficient values of bodies moving.

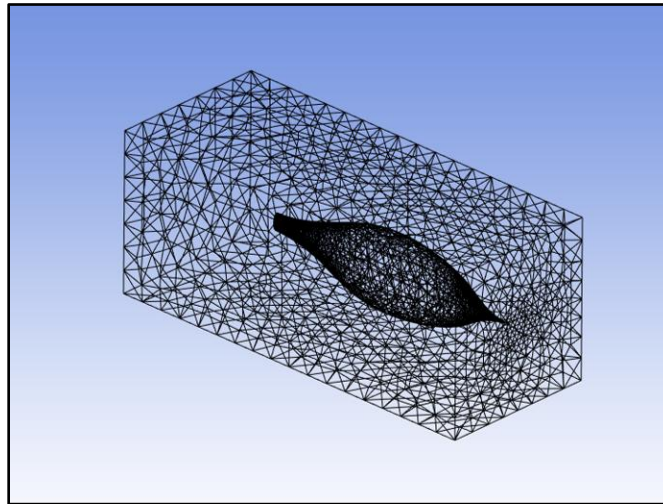


Figure 6. ANSYS Mechanical Mesh Example

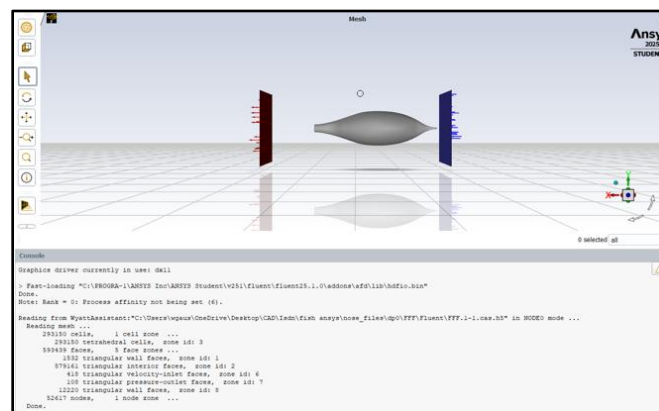


Figure 6. ANSYS Mechanical Mesh Example

Results

Design	Drag Coefficient
Validation Case	0.361392
Benchmark	0.09990803
Sail Fish	0.1341244
Narwhal	0.1021232

Conclusion

The CFD analysis was used to evaluate the effect of external body geometry changes on the hydrodynamic performance of the robotic fish design. A nose cone feature was investigated as a potential method for reducing flow separation and improving drag performance; however, the results showed that the addition of the nose cone did not provide a meaningful reduction in drag coefficient compared to the baseline geometry. A sailfish-inspired body profile was also evaluated due to its naturally streamlined shape, but the results indicated that this design did not provide a significant hydrodynamic advantage while increasing design complexity and limiting internal packaging space. Based on these results, the final design prioritized a balance between hydrodynamic performance, manufacturability, and integration of the internal actuation system.